

UCS2603 THEORY OF COMPUTATION
CAT 2 Answer key

Part – A ($4 \times 2 = 8$ Marks)

1. Define Pushdown automata.
2. Construct Turing machine to find 1's complement of a binary number.

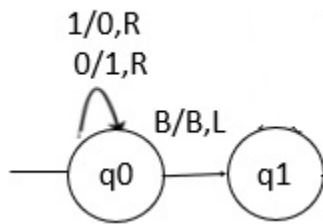
Answer:

Step 1 – Start scanning the input from left to right.

Step 2 – If the R/W is at 1, then make it 0 and move right.

Step 3 – If the R/W is at 0, then make it 1 and move right.

Step 4 – Repeat the steps given above and we will reach B (blank).



3. Explain the Move interpretation of a Turing machine with an example.

Answer:

Moves of a TM M denoted by $\vdash_M$ or $\vdash$ as follows:

If $\delta(q, X_i) = (p, Y, L)$

$X_1X_2\dots X_{i-1}qX_iX_{i+1}\dots X_n \vdash$

$X_1X_2\dots X_{i-2}pX_{i-1}YX_{i+1}\dots X_n$

If $\delta(q, X_i) = (p, Y, R)$

$X_1X_2\dots X_{i-1}qX_iX_{i+1}\dots X_n \vdash$

$X_1X_2\dots X_{i-2}X_{i-1}YpX_{i+1}\dots X_n$

4. Identify the languages under decidable and undecidable problems.

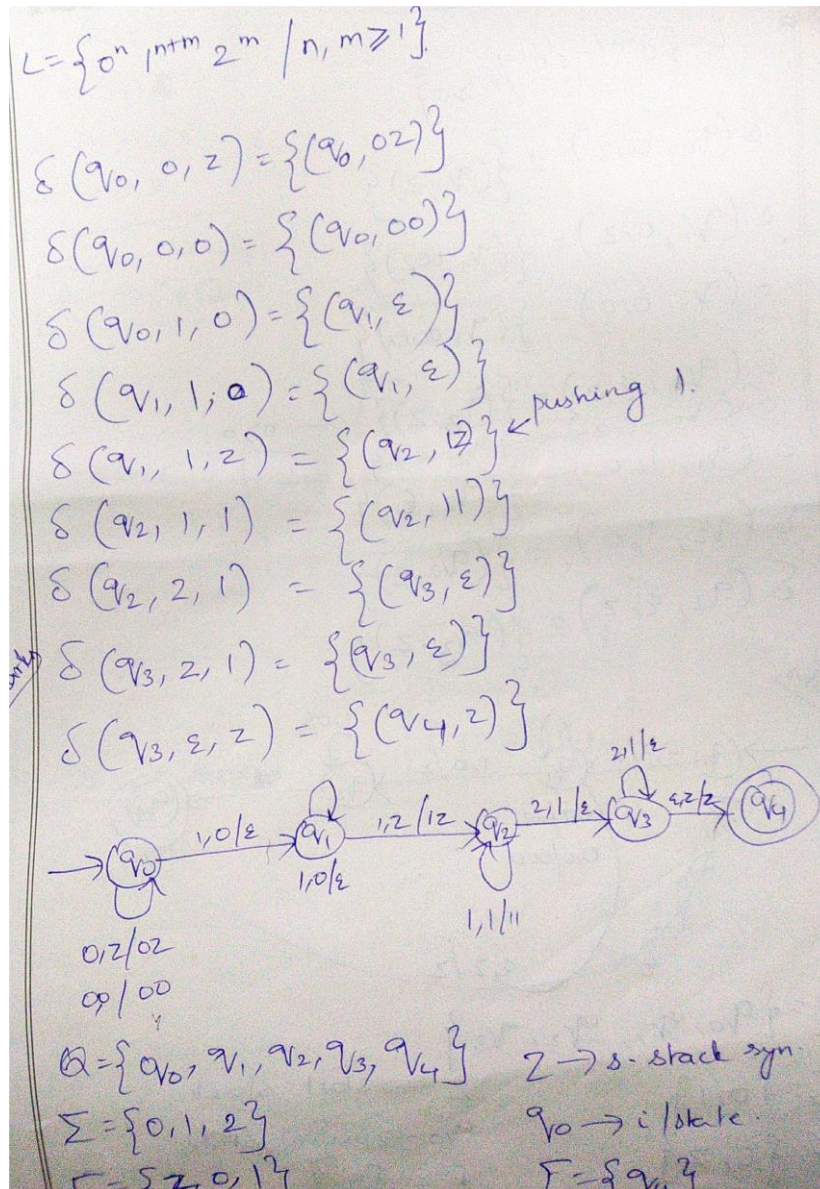
Regular language $\rightarrow D$
 Recursive $\rightarrow D$
 CFL $\rightarrow D$
 Non RE lan $\rightarrow UD$
 CSL $\rightarrow D$
 RE lan $\rightarrow UD$

Part – B (3×6 = 18 Marks)

5. Construct a PDA for the following language.

$$L = \{0^n 1^{n+m} 2^m \mid m, n \geq 1\}$$

Show the validation of $w = 011122$.



b) string validation using id

- Apply pumping lemma for Context Free Languages (CFL) to find the language $L = \{x^n y^n z^n \mid n \geq 1\}$ is CFL or not.
- Explain the closure properties of Recursive(R) and Recursively Enumerable (RE) languages with appropriate examples.

Recursive language.

- i) Recursive languages are closed under complementation.



- (ii) Recursive languages are closed under union.

- (iii) " " " " " intersection.

Recursive enumerable languages.

- i) RE are not closed under complementation.

- (ii) RE are closed under union

- (iii) " " " " " intersection.

|| $w = u$

Part – C (2 × 12 = 24 Marks)

8. Construct a Chomsky Normal Form (CNF) representation for the given Context Free Grammar.

$$S \rightarrow ASB \mid \epsilon$$

$$A \rightarrow aAS \mid a$$

$$B \rightarrow SbS \mid A \mid bb$$

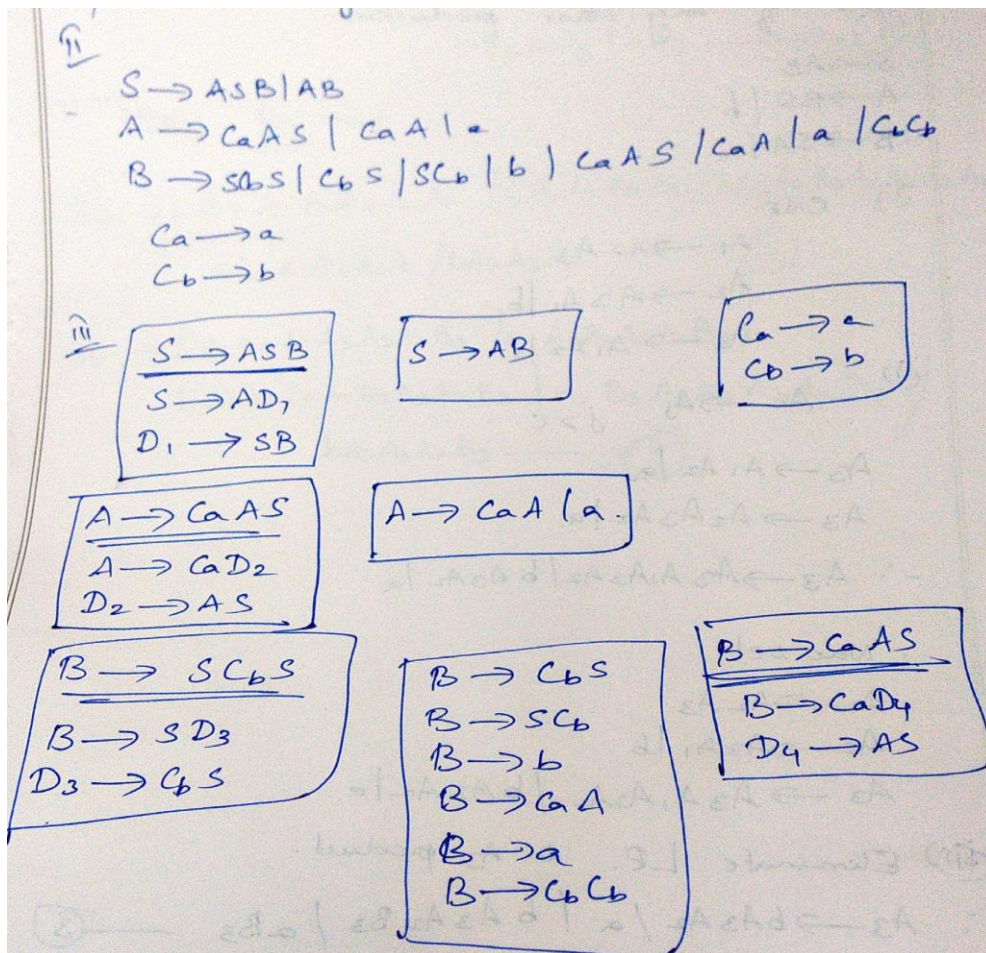
Ans:

$$\begin{aligned} S &\rightarrow ASB \mid \epsilon \\ A &\rightarrow aAS \mid a \\ B &\rightarrow SbS \mid A \mid bb \end{aligned}$$

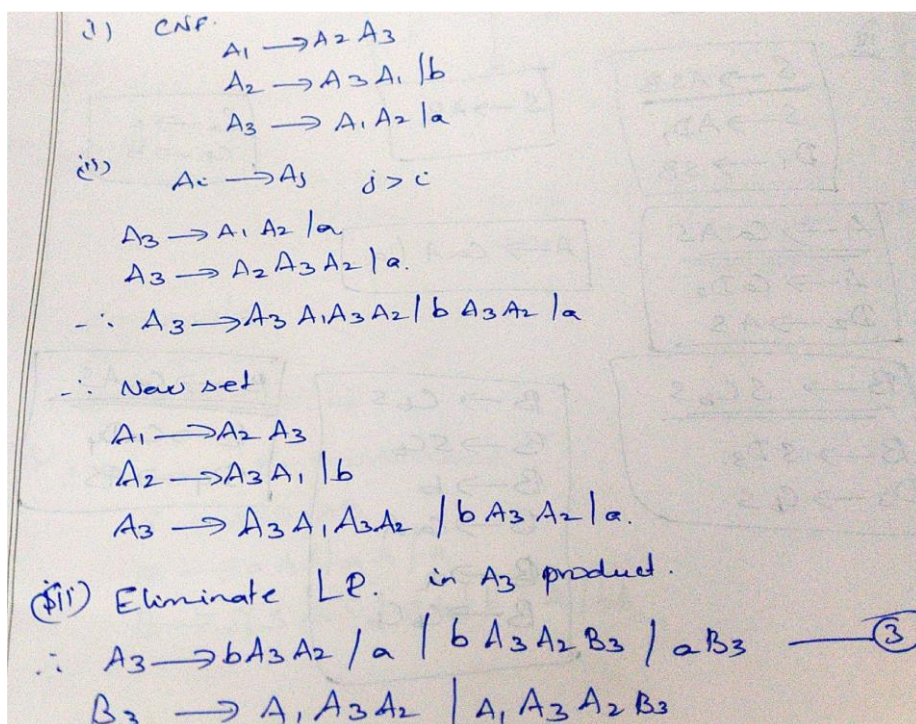
✓ i) Remove ϵ -productions

$$\begin{aligned} S &\rightarrow ASB \mid AB \\ A &\rightarrow aAS \mid aA \mid a \\ B &\rightarrow SbS \mid bS \mid Sb \mid b \mid A \mid bb \end{aligned}$$

$$\begin{aligned} S &\rightarrow ASB \mid AB \\ A &\rightarrow aAS \mid aA \mid a \\ B &\rightarrow SbS \mid bS \mid Sb \mid b \mid aAS \mid aA \mid a \mid bb \end{aligned}$$



9. Construct a Greibach Normal Form (GNF) representation for the given Context Free Grammar.
- $$S \rightarrow AB$$
- $$A \rightarrow BS \mid b$$
- $$B \rightarrow SA \mid a$$



Modify Ai productions

$$\begin{aligned}
 A_2 &\rightarrow b A_3 A_2 A_1 / a A_1 / b A_3 A_2 B_3 A_1 / a B_3 A_1 / b \\
 A_1 &\rightarrow b A_3 A_2 A_1 A_3 / a A_1 A_3 / b A_3 A_2 B_3 A_1 A_3 / \\
 &\quad a B_3 A_1 A_3 / b A_3 \quad \text{--- (1)} \\
 \text{iv) Modify } B_n &\rightarrow a? \\
 B_3 &\rightarrow b A_3 A_2 A_1 A_3 A_3 A_2 / a A_1 A_3 A_3 A_2 / b A_3 A_2 B_3 A_1 A_3 A_3 A_2 / \\
 &\quad a B_3 A_1 A_3 A_3 A_2 / b A_3 A_3 A_2 \quad \text{--- (2)} \\
 B_3 &\rightarrow b A_3 A_2 A_1 A_3 A_3 A_2 B_3 / a A_1 A_3 A_3 A_2 B_3 / \\
 &\quad b A_3 A_2 A_3 A_1 A_3 A_3 A_2 B_3 / a B_3 A_1 A_3 A_3 A_2 B_3 / \\
 &\quad b A_3 A_3 A_2 B_3 \quad \text{--- (5)}
 \end{aligned}$$

10. (a). Construct a Turing Machine for the following language

$$L = \{0^n 1^n 2^n \mid n \geq 1\}$$

- (b). Show the validation of $w = 001122$ and $w = 0012$

Forward

$$\begin{aligned}
 \delta(q_0, 0) &= (q_1, x, R) & \delta(q_3, 0) &= (q_3, 0, L) \\
 \delta(q_1, 0) &= (q_1, 0, R) & \delta(q_3, x) &= (q_0, x, R) \\
 \delta(q_1, 1) &= (q_2, y, R) \\
 \delta(q_1, y) &= (q_1, y, R) \\
 \delta(q_2, 1) &= (q_2, 1, R) \\
 \delta(q_2, z) &= (q_2, z, R) \\
 \delta(q_2, 2) &= (q_3, z, L)
 \end{aligned}$$

Termination

$$\begin{aligned}
 \delta(q_0, y) &= (q_4, y, R) \\
 \delta(q_4, y) &= (q_4, y, R) \\
 \delta(q_4, z) &= (q_5, z, R) \\
 \delta(q_5, z) &= (q_5, z, R) \\
 \delta(q_5, B) &= (q_6, B, R)
 \end{aligned}$$

Backward

$$\begin{aligned}
 \delta(q_3, z) &= (q_3, z, L) \\
 \delta(q_3, 1) &= (q_3, 1, L) \\
 \delta(q_3, y) &= (q_3, y, L)
 \end{aligned}$$

Validate

001122 0012

11. (a). Construct a Turing Machine to subtract 2 numbers.

$$f(m, n) = m - n \text{ if } m > n$$

$$f(m, n) = B \text{ if } m \leq n$$

- (b). Find the solution for **2-1** using the constructed Turing Machine

The function ∂ is described below.

$$\partial(q_0, 0) = (q_1, B, R) \text{ Begin. Replace the leading 0 by B.}$$

$$\partial(q_1, 0) = (q_1, 0, R) \text{ Search right looking for the first 1.}$$

$$\partial(q_1, 1) = (q_2, 1, R)$$

$\partial(q_2,1) = (q_2,1,R)$ Search right past 1's until encountering a 0. Change that 0 to 1.
 $\partial(q_2,0) = (q_3,1,L)$

$\partial(q_3,0) = (q_3,0,L)$ Move left to a blank. Enter state q_0 to repeat the cycle.

$\partial(q_3,1) = (q_3,1,L)$

$\partial(q_3,B) = (q_0,B,R)$

If in state q_2 a B is encountered before a 0, we have situation i described above. Enter state q_4 and move left, changing all 1's to B's until encountering a B. This B is changed back to a 0, state q_6 is entered and M halts.

$\partial(q_2,B) = (q_4,B,L)$

$\partial(q_4,1) = (q_4,B,L)$

$\partial(q_4,0) = (q_4,0,L)$

$\partial(q_4,B) = (q_6,0,R)$

If in state q_0 a 1 is encountered instead of a 0, the first block of 0's has been exhausted, as in situation (ii) above. M enters state q_5 to erase the rest of the tape, then enters q_6 and halts.

$\partial(q_0,1) = (q_5,B,R)$

$\partial(q_5,0) = (q_5,B,R)$

$\partial(q_5,1) = (q_5,B,R)$

$\partial(q_5,B) = (q_6,B,R).$